In [1]:
```
#EXP :- 2
```

In [2]:
```
# Aim :To perform Operation of Exploratory Data Analysis (EDA).
```

In [1]:
```
#Name:Arya patil
#Roll:42
#sec:C
#subject:DSS
#Date:31-07-2026
```

In [2]:
```
import pandas as pd
```

In [3]:
```
import os
```

In [4]:
```
os.getcwd()                                    # Get current working di
```

Out[4]:
```
'C:\\Users\\aryap'
```

In [6]:
```
os.chdir("C:\\Users\\aryap\\OneDrive\\Desktop")              # change dire
```

In [7]:
```
data=pd.read_csv("heart.csv")
```

In [9]:
```
data.head()                                    # first 5
```

Out[9]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	targe
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	

In [10…
```
data.tail()                                    # last 5
```

Out[10]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	ta
298	57	0	0	140	241	0	1	123	1	0.2	1	0	3	
299	45	1	3	110	264	0	1	132	0	1.2	1	0	3	
300	68	1	0	144	193	1	1	141	0	3.4	1	2	3	
301	57	1	0	130	131	0	1	115	1	1.2	1	1	3	
302	57	0	1	130	236	0	0	174	0	0.0	1	1	2	

In [11…
```
data.shape                                     # here (cloumns , rows)
```

Out[11]:
```
(303, 14)
```

In [12…
```
data.columns
```

Out[12]:
```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thala
ch',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],
      dtype='object')
```

In [13…
```
data.info
```

Out[13]:
```
<bound method DataFrame.info of      age  sex  cp  trestbps  chol  fbs
restecg  thalach  exang  oldpeak  \
0     63    1   3       145   233    1        0      150      0
2.3
1     37    1   2       130   250    0        1      187      0
3.5
2     41    0   1       130   204    0        0      172      0
1.4
3     56    1   1       120   236    0        1      178      0
0.8
4     57    0   0       120   354    0        1      163      1
0.6
..   ...  ...  ..       ...   ...  ...      ...      ...    ...
...
298   57    0   0       140   241    0        1      123      1
0.2
299   45    1   3       110   264    0        1      132      0
1.2
300   68    1   0       144   193    1        1      141      0
3.4
301   57    1   0       130   131    0        1      115      1
1.2
302   57    0   1       130   236    0        0      174      0
0.0

     slope  ca  thal  target
0        0   0     1       1
1        0   0     2       1
2        2   0     2       1
3        2   0     2       1
4        2   0     2       1
..     ...  ..   ...     ...
298      1   0     3       0
299      1   0     3       0
300      1   2     3       0
301      1   1     3       0
302      1   1     2       0

[303 rows x 14 columns]>
```

In [14…
```
data.dtypes
```

Out[14]:
```
age           int64
sex           int64
cp            int64
trestbps      int64
chol          int64
fbs           int64
restecg       int64
thalach       int64
exang         int64
oldpeak     float64
slope         int64
ca            int64
thal          int64
target        int64
dtype: object
```

In [15…
```
data.size                                    # Row * columns
```

Out[15]:
```
4242
```

In [16…
```
data.ndim                                    # no. of dimension
```

Out[16]:
```
2
```

In [17…
```
data.describe()
```

Out[17]:

	age	sex	cp	trestbps	chol	fbs	restecg	
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	3
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528053	1
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525860	
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	1
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	1
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000	1
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	2

In [ ]:

In [ ]:

In [ ]: